

Round Table

1. a. Find how many ways five people can be arranged in:
 - i. a line
 - ii. a circle
 b. Find how many ways ten people can be arranged in:
 - i. a line
 - ii. a circle
2. Eight people are arranged in:
 - a. A straight line
 - b. A circle
 In how many ways can they be arranged so that two particular people sit together?
3. Bob, Betty, Ben, Brad and Belinda are to be seated at a round table. Find how many ways this can be done:
 - a. If there are no restrictions
 - b. If Bob is to sit in his favourite chair
 - c. If Betty sits on Bob's right-hand side
 - d. If Brad is to sit between Bob and Ben
 - e. If Belinda and Betty are not to sit next to one another
4. Four boys and four girls are arranged in a circle. Find how many ways this can be done:
 - a. If there are no restrictions
 - b. If the boys and the girls alternate
 - c. If the boys and girls are in distinct groups
 - d. If a particular boy and girl wish to sit next to one another
 - e. If two particular boys do not wish to sit next to one another
 - f. If one particular boy wants to sit between two particular girls
5. The letters A, E, I, P, Q and R are arranged in a circle. Find the probability that:
 - a. The vowels are together
 - b. A is opposite R
 - c. The vowels and consonants alternate
 - d. At least two vowels are next to one another
6. Find how many ways the integers 1, 2, 3, 4, 5, 6, 7, 8 can be placed in a circle if:
 - a. There are no restrictions
 - b. All the even numbers are together
 - c. The odd and even numbers alternate
 - d. At least three odd numbers are together
 - e. The two numbers 1 and 7 are next to one another
7. A committee of three women and seven men is to be seated randomly at a round table
 - a. What is the probability that the three females will sit together?
 - b. The committee elects a president and a vice-president. What is the probability that they are sitting opposite one another?
8. Find how many arrangements of n people around a circle are possible if:
 - a. There are no restrictions
 - b. Two particular people must sit together
 - c. Two particular people sit apart
 - d. Three particular people sit together
9. Twelve marbles are to be placed in a circle. Find how many ways can this be done if:
 - a. All the marbles are of different colours
 - b. There are eight red, three blue and one green marble
10. There are two distinct round tables, each with five seats. In how many ways may a group of ten people be seated?
11. A sports committee consisting of four rowers, three basketballers and two cricketers sits at a circular table.
 - a. How many different arrangements of the committee are possible if the rowers and basketballers both sit together in groups, but no rower sits next to a basketballer?

b. One rower and one cricketer are related. If the conditions in (a) apply, what is the probability that these two members of the committee will sit next to one another?

12. A group of n men and $n + 1$ women sit around a circular table. How many arrangements are possible if no two men are to sit next to one another?

Answer:

1. a. i. 120, ii. 24; b. i. 3,628,800, ii. 362,880

2. a. 10,080; b. 1440

3. a. 24; b. 24; c. 6; d. 4; e. 12

4. a. 5040; b. 144; c. 576; d. 1440; e. 3600; f. 240

5. a. $\frac{3}{10}$; b. $\frac{1}{5}$; c. $\frac{1}{10}$; d. $\frac{9}{10}$

6. a. 5040; b. 576; c. 144; d. 2304; e. 1440

7. a. $\frac{1}{12}$; b. $\frac{1}{9}$

8. a. $(n - 1)!$; b. $2 \times (n - 2)!$; c. $(n - 3) \times (n - 2)!$; d. $6 \times (n - 3)!$

9. a. 39,916,800; b. 165

10. 145,152

11. a. 288; b. $\frac{1}{4}$

12. $\frac{n!(n+1)!}{(2n)!}$